Fraction Operations

Concept

Fractions name equal parts of a whole. Choose the operation before calculating.

Add / subtract

Use a common denominator; combine only numerators.

Multiply

Multiply numerators and denominators; cancel common factors.

Divide

Multiply by the reciprocal of the non-zero divisor.

Order

Brackets → powers → multiply/divide → add/subtract.

Example 1

Calculate $-2/3 + 5/8$.

Solution

1. Common denominator: 24.

2. $-2/3 + 5/8 = -16/24 + 15/24 = -1/24$.

Fraction Operations

Example 2

Calculate $1\ 1/2 \div 2\ 1/4$.

Solution

1. Convert mixed numbers: $3/2 \div 9/4$.

2. Multiply by reciprocal: $(3/2)(4/9) = 2/3$.

Example 3

Calculate $3/4 - (1/2)(2/3)$.

Solution

1. Multiply first: $(1/2)(2/3) = 1/3$.

2. Then subtract: $3/4 - 1/3 = 9/12 - 4/12 = 5/12$.

Fraction Operations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. $7/10 + 1/6$.

2. $-3/4 - 5/8$.

3. $(4/9)(-3/8)$.

4. $5/6 \div (-10/9)$.

5. $2 - (3/4)^2$.

6. A 3 1/2 m ribbon is cut into 1/4 m lengths. How many full pieces?

Fraction Operations

Practice answers | check after trying

1. 21/30 + 5/30 = 26/30 = 13/15.

2. -6/8 - 5/8 = -11/8.

3. -12/72 = -1/6.

4. (5/6)(-9/10) = -3/4.

5. 32/16 - 9/16 = 23/16.

6. (7/2) ÷ (1/4) = 14 pieces. Check: 14 × 1/4 = 3 1/2 m.